V-L-61d
February 20, 1963
SUPERSEDING
Fed. Spec. V-L-61c
October 20, 1959

FEDERAL SPECIFICATION

LACES, FOOTWEAR, NYLON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of nylon laces for use as closures for various types of footwear.

1.2 Classification. Laces shall be of two types in the following classes, as specified (6.2).

Type I.—Spun nylon laces.

Class 1.—7/32 inch width, flat.
Class 2.—10/32 inch width, flat.

Type II.—Bulked filament nylon laces.

Class 1.—3/32 inch diameter, round.
Class 2.—8/32 inch width, flat.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-268—Paper, Kraft, Untreated, Wrapping.
UU-T-111—Tape; Paper, Gummed (Sealing and Securing).
CCC-T-191—Textile Test Methods.
PPP-B-576—Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-T-60—Tape; Pressure Sensitive Adhesive, Waterproof—for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Specification:

MIL-L-10547—Liners, Case, Waterproof.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Sample. Samples are furnished solely for guidance and information to the contractor (6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

3.2 Preproduction sample. When specified (6.2), before production is commenced, the contractor shall submit a finished pair of laces to the contracting officer for approval.

3.3 Material.

3.3.1 *Nylon.* The nylon used in the manufacture of the laces shall be a type of nylon of sufficient characteristics to conform to the requirements of this specification. The use of regenerated or reprocessed nylon is prohibited. The contractor shall certify that the laces have been manufactured from virgin nylon.

3.3.2 *Yarn.* The nylon yarn for braiding type I, classes 1 and 2 laces shall be spun from nylon staple and twisted into a 2-ply yarn. The nylon yarn for braiding type II, classes 1 and 2 laces, shall be a 210-denier filament yarn which has been increased in size to 230-250 denier by air bulking.

3.3.3 *Tips.* The tips shall be made from nylon sheet or film having a thickness of .013 to .016 inch. The color of the tips shall be black for black laces, and clear for all other colored laces.

3.4 Color. The color of the finished laces shall be as specified (6.2).

3.4.1 *Color matching.* The color shall match the standard sample under natural (north sky) daylight, or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.4.2 *Colorfastness.*

3.4.2.1 *Rating per standard sample.* The dyed laces shall show fastness to water equal to, or better than, the standard sample when tested as specified in 4.3.2.2.

3.4.2.2. When no standard sample is available, the dyed laces shall show good fastness to water when tested as specified in 4.3.2.2.

3.5 Braid. The braid for the laces shall be tubular braided without a core using a two over and two under braiding, except type II, class 2 which shall braid one pair over and one pair under. The braid shall be pressed flat for type I, classes 1 and 2, and for type II, class 2 laces in accordance with commercial practice. Type II, class 1 braid for laces shall not be pressed and shall be round. The braid used in the lace shall conform to the requirements in table I when tested as specified in 4.3.2.

3.6 Tipping. Both ends of the braid shall be provided with a firmly attached and secured tip made from the material specified in 3.3.3. Tips shall be 3/4 ± 1/16 inch long.

3.6.1 *Gripping strength.* The nylon tips shall have a minimum gripping strength of 25 pounds for class 1 laces and 40 pounds for class 2 laces when subjected to the test specified in 4.3.2.2.

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TABLE I.- Physical requirements

Type and class	Width Tol.± 1/32 inch	Ends per carrier	Carriers minimum	Picks per inch minimum	Breaking strength minimum
Type I, class 1	7/32	1	44	38	95
Type I, class 2	10/32	2	44	26	175
Type II, class 1	3/32 (diameter)	3	16	28	100
Type II, class 2	8/32	2	44*	17	190

*Carriers braiding in pairs.

3.6.2 *Compression strength.* The tips shall not be cracked, dislodged from their position, or peeled from the lace when tested as specified in 4.3.2.2.

3.7 **Length.** Laces shall be furnished in lengths, including tips, as specified (6.2). The laces shall be cut so as to insure the specified length when tested as specified in 4.3.2.2. Laces under 40 inches may have a length tolerance of plus 5 percent, and laces 40 inches and longer may have a length tolerance of plus 3 percent.

3.8 **Workmanship.** The finished laces shall conform to the quality and grade of product established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Contractor's inspection responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Preacceptance inspection.** The preproduction sample submitted in accordance with 3.2 shall be inspected and tested for all provisions of this specification applicable to end item examinations and tests.

4.3 **Inspection for acceptance.** Inspection shall be in accordance with the provisions of MIL-STD-105 except when otherwise indicated.

4.3.1 *Testing of components.* In addition to the quality assurance provisions of subsidiary specifications and drawings, testing shall be performed on the components listed in table II for the characteristics shown therein. The inspection level for each component shall be level L-1 and the acceptable quality level (AQL) for each component shall be 6.5 test failures per hundred units. All requirements are applicable to the sample unit. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in table II, they shall be as required by the referenced test method.

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TABLE II.- Component tests

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as	Sample unit
Nylon sheet (10 sheets)	Thickness	3.3.3	Gauge	3	Average of 3 determinations to the nearest 0.001 inch	6 by 1½ inch piece or equivalent
	Material identification	3.3.3	Standard Commercial	1	Pass or fail	
Nylon yarn Type II, before & after air bulking (1 cone, tube or spool of each)	Filament	3.3.2	Visual	1	Pass or fail	1 cone, spool or tube of each
	Denier 1/	3.3.2	4021 of CCC-T-191			

1/ The contractor shall certify that a 210 denier filament yarn was used prior to air bulking.

4.3.2 *Inspection of the end item.*

4.3.2.1 *Examination of the end item.* The laces shall be examined for defects in workmanship, material, overall width or diameter, and overall length of tips. Defects found during these examinations shall be classified in accordance with 4.3.2.1.1, 4.3.2.1.2, and 4.3.2.1.3. The sample unit for these examinations shall be one completely finished lace, and the lot size shall be expressed in units of one lace each.

4.3.2.1.1 *Examination of laces for defects in material and workmanship.* The inspection level for this examination shall be level I of MIL-STD-105 and the acceptable quality level (AQL) shall be 1.0 defects per hundred units.

Examine	Defect
Material	Not fabricated of the specified material.
	Any broken thread.
	Frayed.
	Rub or abrasion marks.
	Hole, cut or tear.
	Unevenly braided resulting in open place or break in continuity of braid.
	Pulled thread or uneven tension resulting in uneven width.
	Other than tubular braided without a core.
	Flat lace, when applicable, not flat for a distance of 2 inches or more.
	Round lace, when applicable, not round for a distance of 2 inches or more.

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Examine	Defect
Material (cont'd)	Slub, knot, or abruptly thickened place which may affect utility. Color other than specified. Visibly off shade, i.e., not within established tolerances. Spot, stain, or other discoloration affecting appearance. Objectionable odor.
Tipping	Any tip missing, damaged, malformed, or insecurely attached. Tipping of braid not neatly accomplished. Not fabricated of nylon sheet. Color — clear or any other color than black for black laces. End of tip projects beyond the braid more than 1/32 inch. End of braid projects beyond the tip end more than 1/32 inch.

4.3.2.1.2 *Examination of laces for overall width or diameter.* Any overall width or diameter, measured in accordance with 4.4.1 that exceeds the tolerance of table I shall be classified a defect. The inspection level for this examination shall be L-6 of MIL-STD-105 and the acceptable quality level (AQL) shall be 1.0 defect per hundred units.

4.3.2.1.3 *Examination for overall length of tip ends.* Any overall length of tip exceeding specified tolerances indicated in 3.6 shall be classified a defect. The inspection level for this examination shall be L-6 of MIL-STD-105 and the acceptable quality level (AQL) shall be 1.0 defect per hundred units.

4.3.2.2 *Testing of the end item.* Testing of the completely fabricated laces shall be performed in accordance with table III and CCC-T-191 for characteristics shown therein. The inspection level shall be L-1 of MIL-STD-105 and the acceptable quality level (AQL) shall be 6.5 test failures per hundred units. The sample unit for tests shall be twenty completely fabricated laces. Requirements are applicable to the sample unit. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in table III, they shall be as required by the referenced test method. The lot size shall be expressed in units of one lace each.

4.3.3 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container, fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L-4 and the AQL shall be 4.0 defects per hundred units.

Examination	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Content	Number of bundles per container is more or less than required. Number of laces per bundle is more or less than required.[1]

[1] One bundle from each shipping container in the sample shall be examined.

4.4 **Tests.**

4.4.1 *Width or diameter.* The width or diameter shall be determined at 3 points at least 9 inches apart by a steel rule cali-

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TABLE III.- End item tests

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as
Nylon identification 1/	3.3.1	1530	--	--
Yarn ply	3.3.2	Visual	1	Number of plies
Staple (type I)	3.3.2	Visual	1	Pass or fail
Colorfastness to water	3.4.2	4630	--	--
Weave	3.5	Visual	1	Pass or fail
Gripping strength (tips)	3.6.1	4.4.5	--	--
Compression strength (tips)	3.6.2	4.4.4	1	Pass or fail
Ends per carrier	Table I	Visual	1	Number of ends
Carrier	Table I	Visual	1	Number of carriers
Picks per inch	Table I	Visual	1	Number of picks
Breaking strength	Table I	4102 and 4.4.3	--	--
Length, overall	3.7	4.4.2	1	To the nearest 1/16 inch

1/ The contractor shall submit a certificate of compliance for these characteristics. The certificate shall be accompanied by actual test, inspection or other verifiable quality data.

brated in thirty seconds of an inch, or by a caliper-type gauge and the results averaged. The measurements shall be taken with the test lace suspended vertically under a tension of 3 ounces for class 1 and 6 ounces for class 2 laces.

4.4.2 *Length.* The finished lace shall not be stretched or tensioned prior to the length test. The finished lace shall be suspended vertically under a tension of 3 ounces for class 1 and 6 ounces for class 2, and then measured from end to end including tips.

4.4.3 *Breaking strength.* The breaking strength of the laces shall be determined as specified in CCC-T-191, method 4102, except that the clamps shall be not less than 2 inches in diameter and the dis-

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tance between the center of the clamps at the start of the test shall be 6 inches ± 1/16 inch. When the lace length is less than 24 inches, a flat-type jaw shall be used in lieu of a drum-type jaw.

4.4.4 *Compression strength.* The tip shall be placed between flat blocks of steel and subjected to a gradually applied crushing load of 1000 pounds.

4.4.5 *Gripping strength of tips.*

4.4.5.1 *Test specimen.* The specimen shall be a one-half length of a shoe lace. (Two determinations may be made from one complete shoe lace.)

4.4.5.2 *Apparatus.*

4.4.5.2.1 *Testing machine.* A testing machine as described in method 5100, except a slotted metal plate as shown in figure 1, shall be used in connection with the top jaw assembly and the distance between the upper jaw (slotted metal plate) and the lower jaws or clamps shall be 4 inches ± 1/16 inch. The lower jaw shall be as described in method 5100 or 4102 of CCC-T-191.

4.4.5.2.2 *Metal plate* (*figure* 1). A rigid metal plate of the approximate dimensions 5 inches long by 1-1/2 inches wide, containing a 4-inch tapered slot, centered widthwise and running lengthwise from an inside dimension of 1/32 inch in width, increasing to a width of 1/4 inch at the outside edge. The slot shall be smoothly finished. It is not required that the plate be of a single piece of metal. The slot may be formed by a clamped or bolted assembly of two metal bars or plates properly spaced apart. The plate shall be supported in a horizontal position in, or on, top jaw of the apparatus.

4.4.5.3 *Procedure.* The lace shall be passed upward, tip first, through the jaw assembly and through the wide end of the slot until the tip clears the plate, and then moved horizontally toward the narrow end of the slot until the end of the tip rests on top of the metal plate. The free end of the specimen (lace) shall be fastened in the lower jaw or clamp. The machine shall be started, with the pawl in contact with the ratchet, and the maximum resistance in displacing the tip shall be recorded. The force shall be applied to the specimen at such a rate that the pulling clamp will travel at a uniform speed of 12 inches ± 0.5 inch per minute. After the tip has been forced from the lace, the resultant force shall be read from the dial, scale, or chart and the value recorded to the nearest pound. If the lace tip slips through the slot without its being removed from the lace or if, for any reason attributed to faulty technique, an individual measurement falls markedly below the average test for the sample unit, such result shall be discarded and another specimen shall be tested. Buckling of the tip material at the base of the tip without the tip being removed from the lace shall not be considered as a failure of the tip.

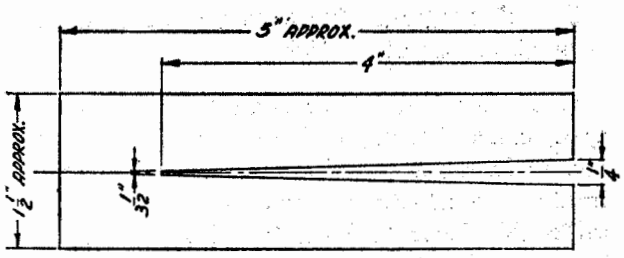


FIGURE 1.—Slotted metal plate.

4.4.5.4 *Report.* Ten specimens shall be tested from each sample unit. The gripping strength of the sample unit shall be the average of the results and shall be reported to the nearest 1.0 pound.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

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5.1 **Packaging.** Packaging shall be level A or C, as specified (6.2).

5.1.1 *Level A.* Five gross of laces of one type, class, length, and color only shall be put-up in a compact bundle having approximate overall dimensions as specified in table IV. Prior to bundling the laces shall be grouped full length in one-gross lot and securely tied together. Each bundle of laces shall be completely wrapped in 60-pound-basis-weight kraft paper conforming to grade B of UU-P-268. The wrap shall be securely sealed with 2-inch minimum-width gummed paper tape conforming to class 2 of UU-T-111.

5.1.2 *Level C.* Laces shall be packaged in accordance with the industry's practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (6.2).

5.2.1 *Level A.* Laces of one type, class, length, and color only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to class 2, V3c or V3s, style RSC of PPP-B-636. Level A packages shall be packed in quantities and arranged within shipping containers having dimensions as specified in table IV. Each fiberboard container shall be provided with a type I or II, grade B or C, class 2 sealed case liner conforming to MIL-L-10547. In lieu of the waterproof case liner, fiberboard containers shall have all seams, corners and manufacturer's joint sealed by the application of 3-inch minimum-width olive-drab color cloth tape conforming to type III, class 1 of PPP-T-60. Alternatively, shipping containers may be used that conform to overseas type of PPP-B-591 or PPP-B-601; style A or B, class 2, of PPP-B-576; or class 2, style 2 or 4 of PPP-B-621, except that the requirements for additional battens shall not apply. The alternative containers shall also be provided with a sealed case liner as specified for the fiberboard container. Each shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the applicable container specification.

5.2.2 *Level B.* Laces of one type, class, length and color only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-636. The minimum bursting strength of the fiberboard shall be 275 p.s.i. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive. Level A packages shall be packed in quantities and arranged within shipping containers having dimensions as specified in table IV. Alternatively, shipping containers may be used that conform to domestic type, style A or B of PPP-B-591 or PPP-B-601; style A or B, class 1, of PPP-B-576; or class 1, style 2 or 4 of PPP-B-621, except that the requirements for additional battens shall not apply.

5.2.3 *Level C.* Laces, packaged as specified in 5.1., shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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TABLE IV. - Packing details

Class	Length	Bundle 1/	Arrangement in container				Shipping container 1/ Inside dimensions (inches)		
		Overall dimensions	Quantity	Length	Width	Depth	Length	Width	Depth
	Inches	Inches							
1	21	10½ by 7½ by 3½	16	2	2	4	21	15	14½
1	25	12½ by 7½ by 3½	16	2	2	4	27	15	14½
1	27	13½ by 7½ by 3½	16	2	2	4	27	15	14½
1	28½	14½ by 7½ by 3½	16	2	2	4	30	15	14½
1	30	15 by 7½ by 3½	16	2	2	4	30	15	14½
1	40	10 by 15 by 3½	16	2	2	4	30	20	14½
1	50	25 by 7½ by 3½	8	1	2	4	27	15	14½
1	52	26 by 7½ by 3½	8	1	2	4	27	15	14½
1	54	27 by 7½ by 3½	8	1	2	4	27	15	14½
1	84	21 by 15 by 3½	8	2	1	4	30	21	14½
2	30	15 by 7½ by 7	8	2	2	2	30	15	14½
2	40	10 by 15 by 7	8	2	2	2	30	20	14½
2	44	11 by 15 by 7	8	2	2	2	30	22	14½
2	52	26 by 7½ by 7	4	1	2	2	27	15	14½
2	60	30 by 7½ by 7	4	1	2	2	30	15	14½
2	64	16 by 15 by 7	4	2	1	2	30	16	14½

1/ Bundle and container dimensions are approximations and are furnished as a guide only.

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6. NOTES

6.1 **Intended use.** The nylon laces covered by this specification are intended for use on various types of footwear.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and length required (1.2 and 3.7).
(*c*) When preproduction sample is required (3.2).
(*d*) Color required (3.4).
(*e*) Selection of applicable levels of packaging and packing (5.1 and 5.2).

6.3 **Sample.** For access to guide and standard samples, address the procuring office issuing the invitation for bids.

6.4 For civil agencies level B packing should be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

6.5 The requirements in 5.1.1 shall apply when level B packaging is specified for civil agency procurements.

6.6 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Laces, Shoe:
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Laces, braided, shoe:
Truckload minimum weight 18,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY INTERESTS:

Army—Q M

Navy—S MC

Air Force—ASD MAA

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